DS4102 - Assignment 01

Descriptive Data Analysis of the Palmer Penguins Dataset

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1. Introduction

This report presents a descriptive analysis of the Palmer Penguins dataset, which records physical measurements for 344 penguins across three species (Adelie, Chinstrap, and Gentoo) sampled from three islands in the Palmer Archipelago, Antarctica. The dataset contains four numeric variables, bill length, bill depth, flipper length, and body mass, and three categorical variables: species, island, and sex.

The goals of this analysis are to: (1) compute descriptive statistics for each variable, (2) visualize the distribution and behavior of each variable, (3) examine how the numeric variables differ across categories such as species and island, and (4) draw conclusions from the observed patterns.

More details: https://github.com/MalakaSupun/Penguins-EDA-Wizard

2. Dataset Overview

Variables:

1. **species (categorical)** – [Adelie, Chinstrap, Gentoo] | The main grouping factor in the dataset, and the strongest source of variation in every numeric measurement. 152 Adelie (44.2%), 124 Gentoo (36.0%), 68 Chinstrap (19.8%); no missing values.
2. **island (categorical)** - [Biscoe, Dream, Torgersen] | The island in the Palmer Archipelago where the penguin was sampled. 168 Biscoe (48.8%), 124 Dream (36.0%), 52 Torgersen (15.1%). Not an independent factor — each island hosts a different, non-overlapping subset of species, so it behaves largely as a proxy for species rather than a separate geographic effect.
3. **sex (categorical)** - [Male, Female] | Roughly balanced: 168 male (50.5%), 165 female (49.5%), with 11 records missing sex. Within every species, males average heavier body mass than females.
4. **bill_length_mm (numeric) -** Length of the penguin's bill, in millimeters. Mean 43.92 mm, range 32.1–59.6 mm. Adelie has the shortest bills (~38.8 mm); Chinstrap and Gentoo are longer and similar in length (~47.5–48.8 mm).
5. **bill_depth_mm (numeric) -** Depth (vertical thickness) of the bill, in millimeters. Mean 17.15 mm, range 13.1–21.5 mm. Gentoo stands apart with a notably shallower bill (~15.0 mm) than Adelie and Chinstrap (~18.4 mm each)
6. **flipper_length_mm (numeric) -** Length of the flipper, in millimeters. Mean 200.92 mm, range 172–231 mm. Gentoo has markedly longer flippers (~217 mm) than Adelie (~190 mm) and Chinstrap (~196 mm), consistent with its larger overall body size.
7. **body_mass_g (numeric) -** Body mass, in grams. Mean 4201.75 g, range 2700–6300 g. Gentoo is substantially heavier (~5076 g on average) than Adelie (~3701 g) and Chinstrap (~3733 g); within each species, males are heavier than females.

Two rows were missing all four numeric measurements and were excluded from the numeric analysis, leaving 342 usable records. All categorical variables retain their full counts where relevant.

3. Descriptive Statistics: Numeric Variables

Variable	N	Mean	Median	Std Dev	Min	Max	Skew
Bill length (mm)	342	43.92	44.45	5.46	32.1	59.6	0.05
Bill depth (mm)	342	17.15	17.30	1.97	13.1	21.5	-0.14
Flipper length (mm)	342	200.92	197.00	14.06	172.0	231.0	0.35
Body mass (g)	342	4201.75	4050.00	801.95	2700	6300	0.47

All four numeric variables have skewness close to zero (between -0.14 and 0.47), indicating rough symmetric distributions, with a slight right-skew in body mass. Negative kurtosis values (around -0.7 to -1.0) indicate flatter-than-normal distributions, consistent with three distinct species creating multiple humps rather than one sharp peak, as seen in the histograms below.

4. Descriptive Statistics: Categorical Variables

The dataset contains three categorical variables, species, island, and sex, each summarized below by frequency count and percentage of the total sample.

Species	Count	Percent
Adelie	152	44.2%
Gentoo	124	36.0%
Chinstrap	68	19.8%

Species are the most unevenly distributed of the three: Adelie is the most common (152 birds, 44.2%), followed by Gentoo (124, 36.0%) and Chinstrap (68, 19.8%). This imbalance is worth keeping in mind for later comparisons, since group means based on smaller categories (like Chinstrap) rest on fewer observations and carry more sampling variability.

Island	Count	Percent
Biscoe	168	48.8%
Dream	124	36.0%
Torgersen	52	15.1%

Island shows a similar pattern: Biscoe accounts for the largest share of samples (168, 48.8%), followed by Dream (124, 36.0%) and Torgersen (52, 15.1%). As noted earlier, island isn't an independent group. Each island supports only a subset of species. So, this distribution is really a reflection of where each species happens to have been sampled, not a separate geographic pattern.

Sex	Count	Percent
Male	168	50.5%
Female	165	49.5%

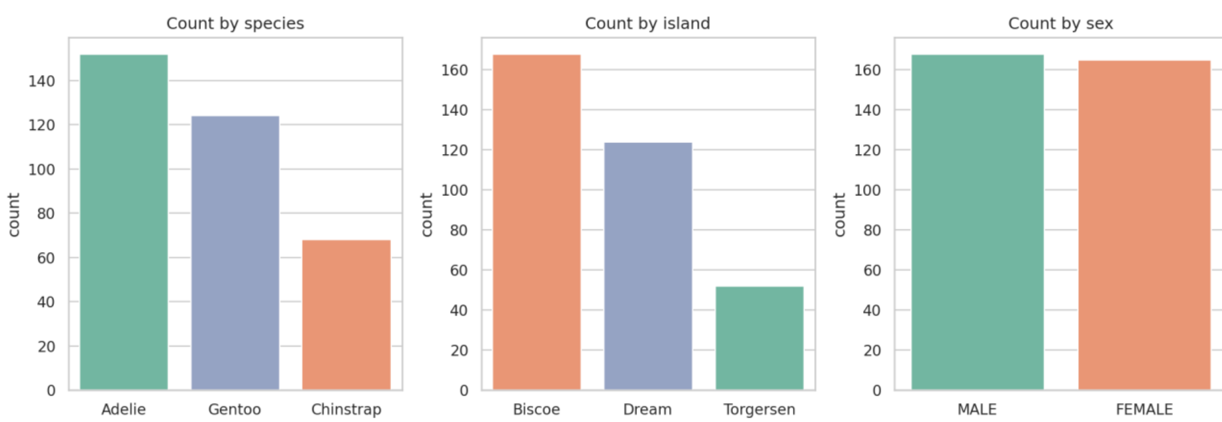

Figure 1. Counts of observations by species, island, and sex.

Adelie is the most common species (44.2%), and Biscoe is the most-sampled island (48.8%). Sex is almost perfectly balanced (50.5% male vs. 49.5% female).

5. Distribution of Numeric Variables

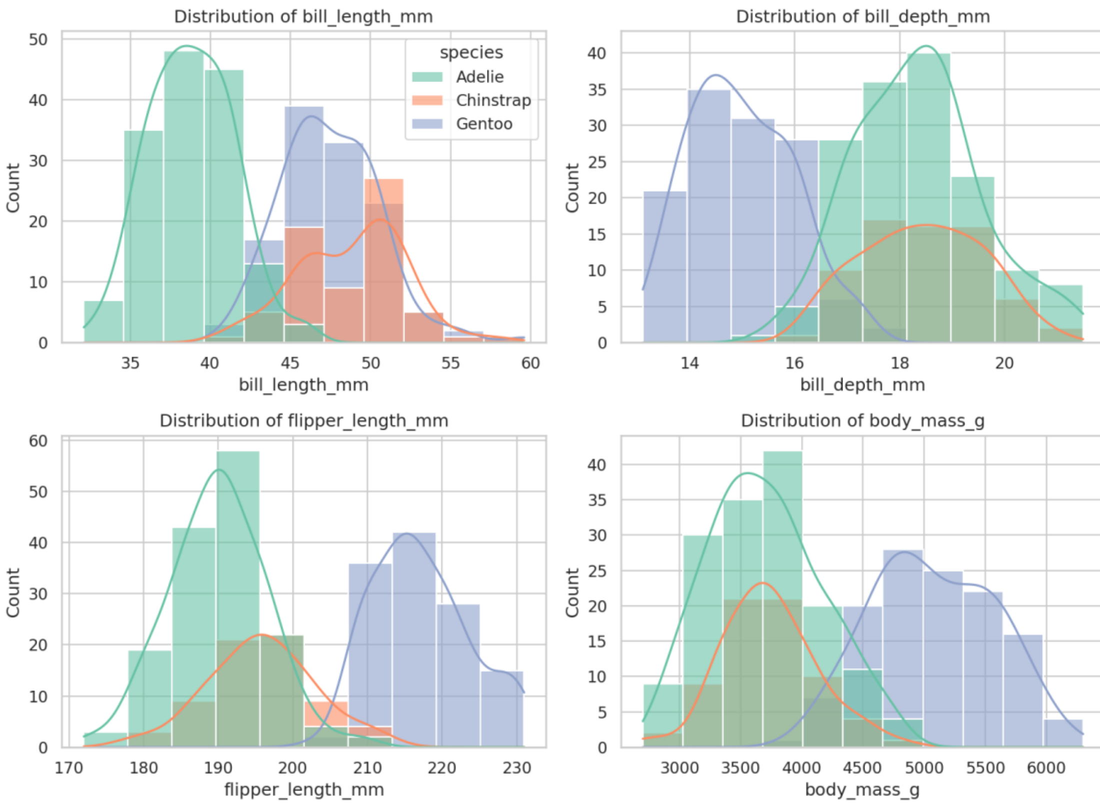

Figure 2. Histograms of the four numeric variables, colored by species.

Each histogram reveals that the apparent single distribution is really a mixture of three species-specific sub-distributions. Bill depth and body mass show the clearest separation, with Gentoo penguins standing apart from Adelie and Chinstrap.

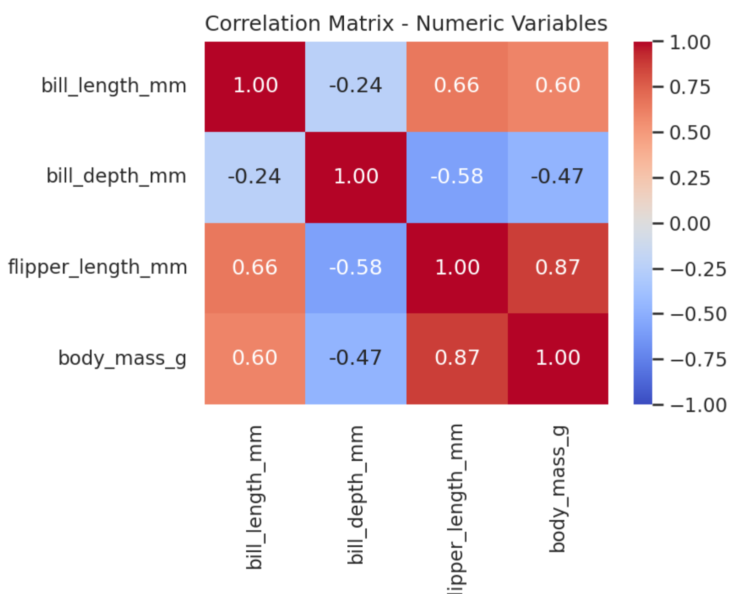

Figure 3. Correlation matrix of numeric variables.

Flipper length and body mass are strongly positively correlated (r ≈ 0.87), as expected for overall body size. Bill depth is negatively correlated with both flipper length (r ≈ -0.58) and body mass (r ≈ -0.47). This is a between-species effect: Gentoo penguins have long flippers, high body mass, and shallow bills, while Adelie/Chinstrap show the opposite pattern.

6. Behavior of Variables by Category

6.1 By Species

Species	Mean bill length	Mean bill depth	Mean flipper length	Mean body mass (g)
Adelie	38.79	18.35	189.95	3700.66
Chinstrap	48.83	18.42	195.82	3733.09
Gentoo	47.50	14.98	217.19	5076.02

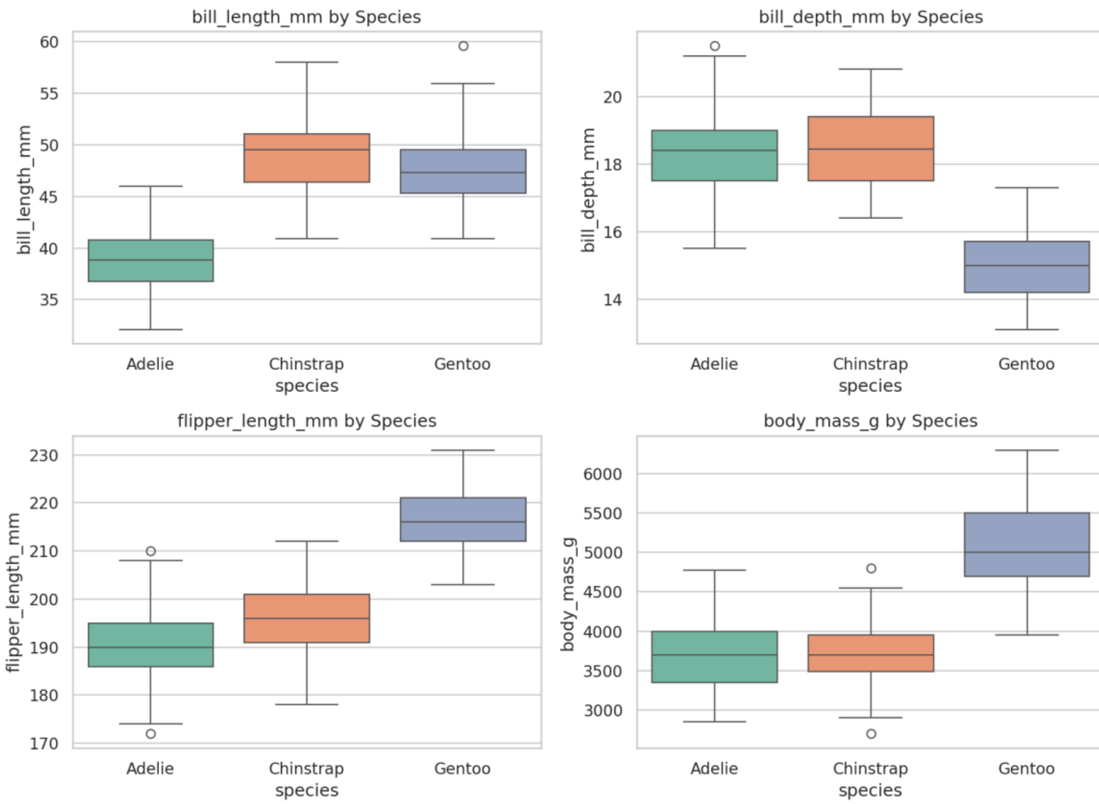

Figure 4. Boxplots of each numeric variable by species.

Species is the dominant source of variation in this dataset. Gentoo penguins are markedly larger (body mass, flipper length) but have notably shallower bills than the other two species. Adelie has the shortest bill length; Chinstrap and Gentoo have similar bill lengths but very different bill depths and body sizes. A one-way ANOVA confirms these differences are statistically significant for body mass ($F = 343.6$, $p < 0.001$) and bill length ($F = 410.6$, $p < 0.001$).

6.2 By Island

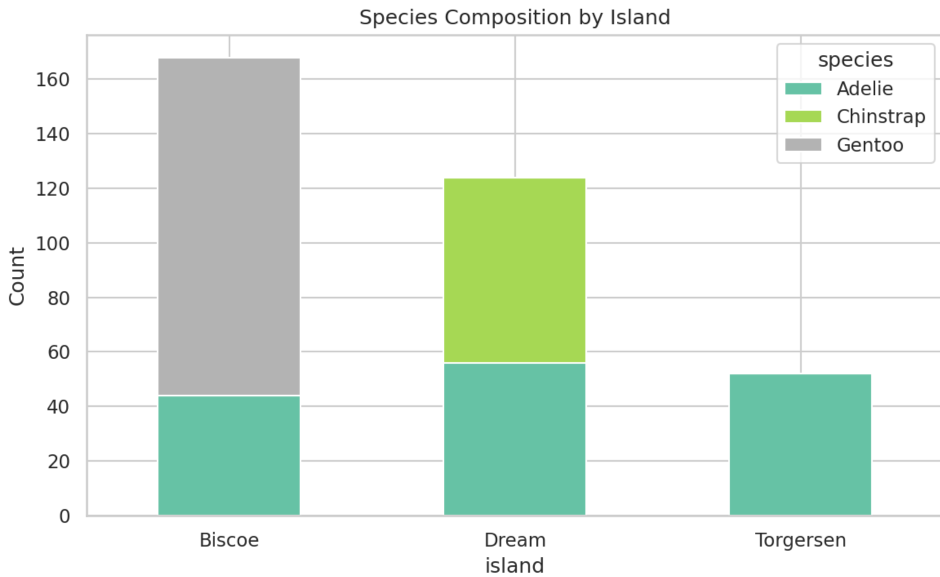

Figure 5. Species composition by island.

Island is not an independent grouping factor. It is almost a proxy for species. Torgersen contains only Adelie penguins, Dream contains Adelie and Chinstrap, and Biscoe contains Adelie and Gentoo. As a result, apparent differences in measurements 'by island' (e.g., higher average body mass on Biscoe) are largely driven by which species happen to live there, rather than an island effect.

6.3 By Sex

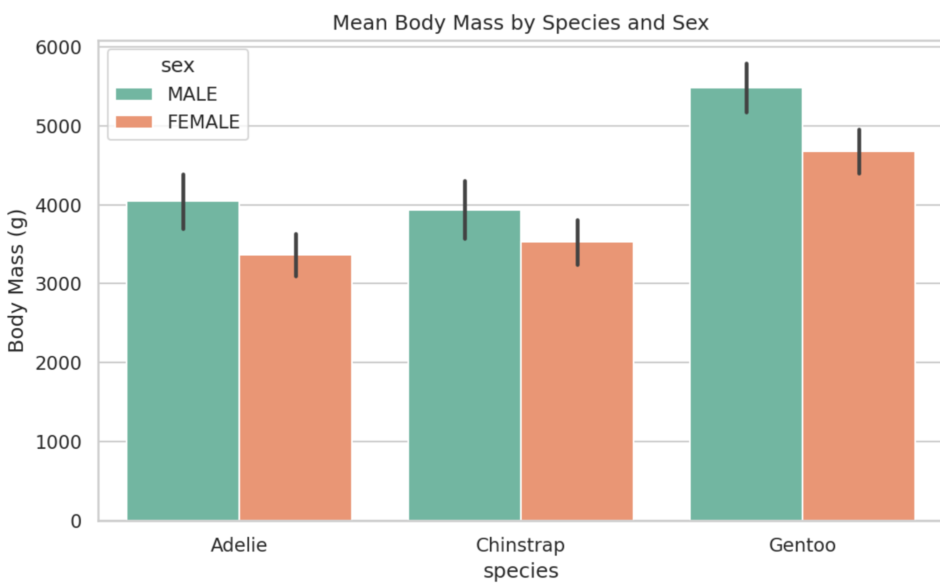

Figure 6. Mean body mass by species and sex, with standard deviation error bars.

Within every species, males are consistently heavier than females. A pattern known as sexual size dimorphism. The gap is present across all three species, meaning sex and species act as independent, additive sources of variation in body mass rather than one masking the other.

6.4 Pairwise Relationships

Figure 7. Pairwise scatterplots and marginal distributions for all numeric variables, colored by species.

The pair plot shows that combinations of just two variables (e.g., bill length and bill depth, or flipper length and bill depth) are enough to separate the three species almost perfectly into distinct clusters. This suggests the four numeric measurements together carry strong discriminative information about species identity.

7. Conclusions

When we look at variations in certain traits, it's clear that the species of penguins are the main influence on these differences. For example, Gentoo penguins stand out as being larger and having shallower bills compared to the others. On the other hand, Adelie penguins have the shortest bills, while Chinstrap penguins fall somewhere in between, although they share some similarities with Gentoo in terms of bill length.

It's important to note that the differences we see among islands are closely tied to the species that inhabit them; each island is home to a unique mix of penguin species. So, when we look at averages from different islands, we're really seeing the impact of which species are present rather than a true geographic effect.

Sex also plays a significant role in their characteristics. Male penguins tend to be heavier than females across all species, showing a consistent pattern known as sexual dimorphism.

In terms of physical traits, there's a strong connection between flipper length and body mass. Larger penguins generally have longer flippers. Interestingly, bill depth behaves differently; it often decreases as overall body size increases, particularly due to Gentoo penguins' relatively shallow bills compared to their larger bodies.

When we analyze the data, we find that the distribution of these measurements tends to be symmetrical but doesn't resemble a typical bell curve; it's a bit flatter than that. This pattern is primarily due to the mix of three different species, rather than any single group that behaves like a well-defined population.

In practical terms, this means that we can accurately predict a penguin's species by looking at just two of the four key measurements, like bill length and bill depth. This insight, backed by the pair plot clustering we observed, opens opportunities for a simple classification exercise to further explore these findings.